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RESEARCH

Adversarially Robust Hardware Trojan Detection with Synthetic Data Augmentation

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Abstract

As semiconductor manufacturing becomes increasingly outsourced to untrusted entities, Hardware Trojan (HT) attacks pose a critical threat to the security and reliability of modern integrated circuits. Machine learning models have improved the effectiveness of HT detection using Ring Oscillator Network (RON) side-channel data, yet recent work shows that these models are highly vulnerable to adversarial attacks. This paper evaluates the robustness of the Support Vector Machine (SVM) classifier, a leading algorithm in state-of-the-art HT detection frameworks, under gradient-based adversarial attacks. The proposed work demonstrates that high nominal accuracy does not ensure security against these attacks, which can reduce recall to zero. To strengthen resilience, three data-augmentation methods are investigated: SMOTE, Conditional Tabular Generative Adversarial Network (CTGAN), and Tabular Variational Autoencoder (TVAE). TVAE produces high-fidelity synthetic samples and substantially improves robustness, maintaining over 91% accuracy for nominal performance and over 88% accuracy under strong adversarial perturbations that cause a 100% attack success rate for the surrogate model. The results highlight the need to reframe hardware security evaluations beyond nominal accuracy toward adversarial robustness.

Keywords Hardware trojan · Adversarial attack · Data augmentation · Side-channel analysis · Data synthesis

1 Introduction

The semiconductor supply chain, characterized by its extensive network of distribution points and increasingly complex system design, presents major risks to integrated circuit (IC) security [1]. A common industry practice involves outsourcing IC fabrication to untrusted or unverified foundries, which has established a major security vulnerability due to possible hardware Trojan (HT) attacks [2]. The stealthy nature of these Trojans together with their tendency to activate only under rare trigger conditions makes them resistant and challenging to detection through conventional functional testing procedures. Successful Trojan attacks can result in severe security breaches that will enable attackers to steal data, block access, and destroy devices.

To address these threats, non-invasive detection methods based on side-channel analysis (SCA) have been widely explored. SCA leverages physical signals that ICs emit unintentionally, such as power, timing, or electromagnetic variations, to detect their operational states and activities. Among these techniques, Ring Oscillator Network (RON)-based detection methods have gained prominence as effective on-chip sensing mechanisms [3]. A RON consists of multiple ring oscillators embedded within the semiconductor chip's power grid, whose operating frequencies are sensitive to power supply fluctuations. When a hardware Trojan activates, its switching activity perturbs the power network, inducing measurable frequency shifts in the ring oscillators. Trojan detection is therefore performed by comparing the frequency differences between chips potentially infected with Trojans and those of known uninfected "golden" chips.

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Despite their promise, RON-based detection methods suffer from limited effectiveness due to process variations and measurement noise, which can obscure Trojan-induced signatures. Early classification techniques, such as Principal Component Analysis (PCA) and convex hull–based methods, achieved high detection accuracy but incurred unacceptably high false positive rates (FPRs), reaching up to 50% [4]. Such high FPRs are particularly problematic in manufacturing settings, as they result in the unnecessary rejection of functional chips and substantial financial losses. Supervised machine learning (ML) techniques have thus been introduced to analyze RON frequency data more effectively. Prior studies have evaluated several classifiers, including K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naive Bayes, and ensemble methods, demonstrating up to a 40% reduction in FPRs while maintaining binary classification accuracy above 90% [5]. These results highlight the potential of supervised ML for improving hardware Trojan detection performance.

However, the adoption of ML-based hardware Trojan detectors introduces a new class of security risks in the form of adversarial attacks. Adversarial machine learning (AML) focuses on crafting subtle perturbations to input data that can mislead or bypass trained ML systems [6]. In security-critical applications, these attacks pose a serious threat, as attackers can exploit such ML model vulnerabilities to induce misclassification with minimal and often imperceptible modifications [7]. Consequently, the high-performing supervised detection models used in hardware security pose a critical challenge because they remain vulnerable to targeted adversarial attacks.

This work systematically investigates how supervised SVM-based hardware Trojan detectors respond to adversarial evasion attacks through a comprehensive analysis of their vulnerability. Specifically, we analyze the impact of adversarial perturbations applied to test data in ML-assisted HT detection settings and demonstrate their effectiveness in degrading detection performance. To mitigate adversarial attacks, we develop an innovative defense strategy through synthetic data augmentation to build an enhanced training dataset for improving model robustness. Our experimental results prove that while supervised ML provides higher hardware Trojan detection performance, such models remain vulnerable to complex and sophisticated adversarial attacks; furthermore, data augmentation emerges as an essential component for strengthening adversarial resilience. Overall, this research demonstrates that robust hardware security requires more than high accuracy; adversarial robustness must be a fundamental consideration in the design of ML-based detection systems.

The main contributions of this work are summarized as follows:

- We show that a state-of-the-art SVM-based RON Trojan detector, despite exceeding 90% nominal accuracy, is catastrophically vulnerable to a gradient-based feature-space evasion attack, with recall collapsing to 0% under a bounded perturbation.
- We formulate a finite-difference gradient evasion attack that applies to non-differentiable kernel SVMs (linear, RBF, and sigmoid), providing the functional equivalent of an FGSM-style attack for hardware side-channel data.
- We propose and evaluate a synthetic-data-augmentation defense, and demonstrate that TVAE-based augmentation restores recall to approximately 88% under attack, outperforming adversarial training, SMOTE, and CTGAN.
- Through a controlled comparison in which SMOTE and TVAE receive identical class balancing, we isolate manifold coverage—rather than class balancing alone—as the source of adversarial robustness, and confirm the stability of this finding across multiple random data splits.

2 Background

2.1 Hardware Trojan and Detection Approaches

The risk of malicious circuit modifications during IC manufacturing has been widely documented in various studies [8, 9]. The malicious modifications in ICs can be classified based on their physical characteristics and their activation and action properties [8]. The detection of stealthy HTs through traditional methods such as physical analysis and functional testing becomes impractical because these methods fail to detect rare activation conditions [10]. The need for non-invasive side-channel analysis (SCA) has become the primary focus for detecting HTs [11]. The fundamental principle of modern hardware security depends on Side-Channel Analysis (SCA) which examines physical signals that devices emit unintentionally [12]. The same principle which attackers use to extract secrets from devices can be used by defenders to detect anomalies [13]. Ring Oscillator Networks (RONs) serve as a particular SCA-based method to detect Trojans through on-chip oscillator frequency changes which indicate power consumption anomalies caused by Trojan switching activity [14]. Recent research has further highlighted the severity of such power network perturbations, demonstrating that hardware Trojans can intentionally induce targeted voltage droops to orchestrate localized timing fault attacks [15]. This phenomenon directly motivates our detection methodology: because malicious payloads fundamentally rely on manipulating these power

and timing behaviors, RONs provide a critical observability window into their physical footprints. However, the reliance on these sensitive side-channel signatures makes securing the downstream machine learning models against adversarial manipulation an urgent priority.

2.2 Threat to Machine Learning

The field of adversarial machine learning (AML) has become essential for cybersecurity because it studies how machine learning models become vulnerable to deliberate attacks [16]. Mainly attackers can perform two types of attacks against ML models: they can add malicious data to the training process (poisoning attacks) or they can make small changes to input data during inference to achieve misclassification (evasion attacks) [17]. A skilled attacker can develop a particular input to defeat the detection system by making a Trojan-infected chip seem harmless [18]. The decision-making process of ML models remains unexplainable to humans because they function as black boxes [19]. A successful adversarial attack on detection systems enables infected chips to pass quality assurance checks and enter the supply chain without detection.

2.3 Threat Model

In this paper, we consider an adversarial threat scenario in which an attacker aims to evade detection by a supervised ML model that identifies hardware Trojans as illustrated in Fig. 1. The adversary operates under a gray-box threat model, assuming partial knowledge of the target system's feature representations and data processing techniques [20]. Specifically, the attacker does not require white-box access to the exact model parameters or underlying architecture; instead, they possess the ability to query the detection system to obtain continuous decision scores, enabling the numerical approximation of gradients required to craft adversarial perturbations. Under this setting, the attacker seeks to craft adversarial examples through side-channel data manipulation of Trojan-infected chips, such that the trained model incorrectly identifies them as non-Trojan. The attack takes place exclusively during the inference phase and does not involve modifying the training process. By introducing carefully designed, small, and imperceptible perturbations to the input data, the adversary ensures that the manipulated side-channel data remain physically plausible and difficult to distinguish from benign variations [21].

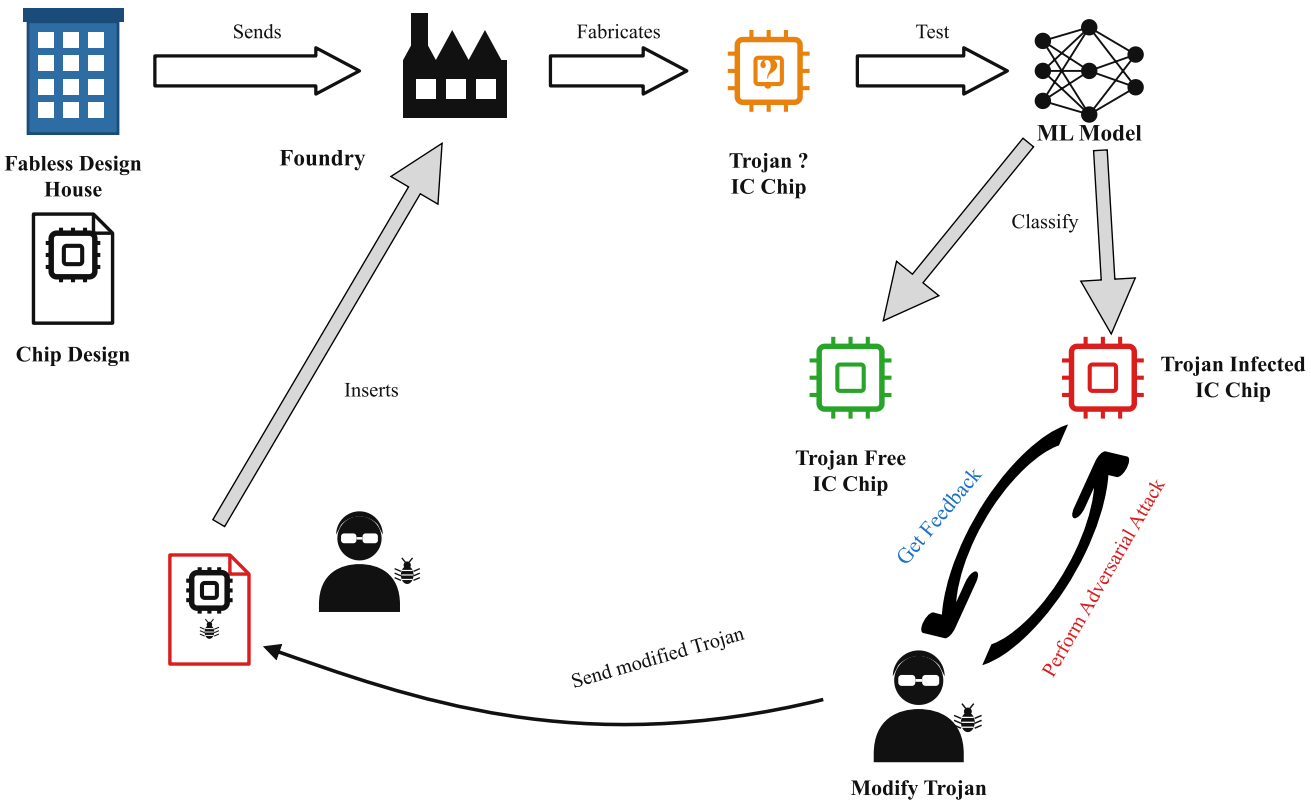


Fig. 1 Overview of the threat model of hardware Trojan modification inserted during the pre-fabrication design stage, and a side-channel–based machine-learning system is employed to detect the intrusion and resist adversarial evasion

The assumption of query access to continuous decision scores is justified by common deployment practice: RON-based screening is frequently exposed as a characterization or debug service that returns a calibration margin or confidence value rather than a bare verdict, and an insider at an untrusted foundry or test house may directly observe such intermediate scores. This assumption is also deliberately conservative. It defines the strongest practically motivated adversary and therefore an upper bound on evasion capability; a weaker attacker who observes only a binary pass/fail outcome can, at most, reconstruct a coarser approximation of the same gradient through additional boundary queries, yielding perturbations no more effective than those studied here. Crucially, our contribution does not depend on the attacker's exact query budget: the proposed defense hardens the geometry of the decision boundary itself, so it improves robustness against any perturbation that pushes a Trojan-inserted sample toward the boundary, independent of how the adversary estimates that direction.

2.4 Related Work

The detection of Hardware Trojans (HT) has evolved from statistical SCA to sophisticated ML techniques capable of modeling complex non-linear relationships in power and frequency data. Early approaches utilized outlier detection and simple classifiers like Support Vector Machines (SVM) to distinguish between infected and golden chips based on path delay and leakage current [22]. More recently, Deep Learning (DL) has been adopted to automate feature extraction. For example, researchers applied Convolutional Neural Networks (CNNs) to classify side-channel waveforms, demonstrating superior accuracy over template attacks [23]. Additionally, contemporary research has introduced advanced spatial and temporal power profiling techniques as well as hybrid machine learning architectures to isolate transient and stealthy Trojan activities [15]. Although these approaches significantly enhance nominal detection performance and feature extraction capability, they largely assume a benign evaluation environment. The proposed framework addresses the limitations of these state of the art timing and power based methodologies. Through a systematic analysis of their structural vulnerabilities, it is demonstrated that, in the absence of explicit defense strategies such as synthetic data augmentation, even highly accurate side channel detectors remain fundamentally susceptible to adversarial evasion. However, these studies predominantly focus on maximizing nominal accuracy in benign environments, operating under the assumption that the test data follows the same distribution as the training data without malicious interference.

As ML models are increasingly deployed in hardware security, their susceptibility to adversarial examples inputs intentionally perturbed to cause misclassification has become a critical concern. While extensively studied in the image domain, this threat surface is equally potent in the hardware domain. Studies demonstrated that ML-based SCA profiling attacks could be defeated by injecting imperceptible noise into power traces, effectively creating adversarial examples that mislead the leakage model [24–26]. Similarly, in the context of Physical Unclonable Functions (PUFs), attackers have utilized adversarial machine learning techniques to model and clone Strong PUF behaviors [27]. These findings indicate that high detection accuracy on clean data does not equate to security; an adversary capable of manipulating environmental conditions (e.g., temperature or voltage noise) can easily evade standard ML-based detectors.

To mitigate evasion attacks, adversarial training (AT), which augments the training process with adversarially perturbed examples, is widely regarded as the standard defense mechanism [28]. However, applying AT to HT detection presents a unique challenge: the available measurements are inherently imbalanced. In the RON benchmark adopted in this study, the Trojan-free (golden) reference measurements form the minority class, being substantially scarcer than the Trojan-inserted samples, as only a limited number of verified golden chips are typically available for characterization. Prior work by Yin et al. [29] highlights that adversarial training under such imbalanced conditions can disproportionately degrade the performance of the minority class, often leading to substantial reductions in recall, as the model overfits to the robust features of the majority class.

This limitation necessitates an alternative approach: synthetic data augmentation (SDA). Generative models, such as Generative Adversarial Networks (GANs), have been explored to synthesize side-channel traces to address data scarcity [30]. By populating the manifold of the minority (Trojan-free) class, generative approaches can restore class balance and enforce smoother decision boundaries. Our work builds upon this by employing TVAE not only to solve the imbalance problem but to demonstrate that a synthetically balanced manifold provides superior intrinsic robustness against gradient-based attacks compared to traditional adversarial learning.

3 Methodology

3.1 Overview

The proposed methodology establishes a systematic framework for evaluating and enhancing the adversarial robustness of HT detection models as illustrated in Fig. 2. The workflow begins with data acquisition, during which

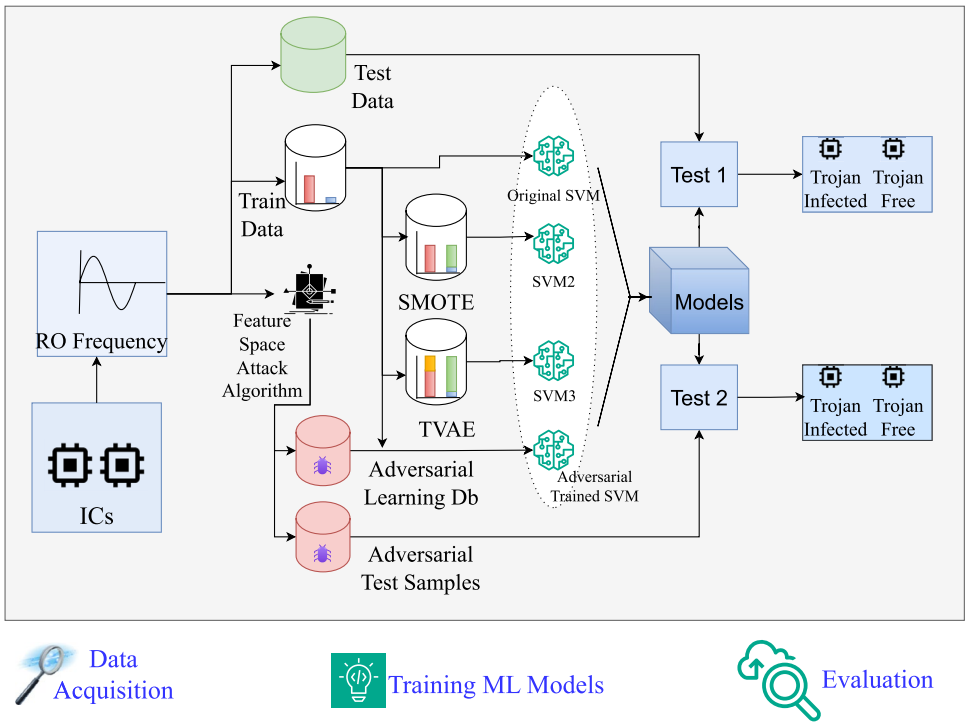


Fig. 2 Overview of proposed framework for evaluation of adversarial attack resilient hardware trojan detection models

side-channel frequency signatures are captured from RON embedded in ICs. Following data partitioning into training and testing sets, the framework employs a feature-space attack algorithm to generate perturbed inputs. The first set of perturbed samples is mixed with the original train dataset to enable adversarial learning during model training. The second set of perturbed samples is held out exclusively to test all the defensive models including data augmentation methods and the baseline SVM to evaluate the robustness against adversarial attack scenario.

The core training phase investigates four distinct model configurations to benchmark resilience: (1) the original SVM, (2) SMOTE-augmented SVM, (3) TVAE-augmented SVM, and (4) adversarially trained SVM. Finally, the evaluation phase subjects these models to a dual-testing protocol (Test 1 and Test 2), which assesses detection performance on both clean test data and adversarial test samples, respectively. This evaluation strategy enables a comprehensive analysis of the trade-off between nominal accuracy and security against evasion attacks.

3.2 Dataset Collection

The dataset used in this study involves eight FPGA boards, each of which is divided into four sections to increase the number of samples while treating each section as an independent IC. Each IC contains frequency measurements from eight 41-stage Ring Oscillators (ROs), which run in each IC under the labels RO1 through RO8. In total, the dataset comprises 800 entries with nine attributes, including eight RO frequency measurement features and a binary classification label indicating whether a chip is Trojan-free (0) or Trojan-infected (1). The utilization of this specific dataset size is a deliberate methodological choice. To rigorously demonstrate the vulnerability of existing detection frameworks, we intentionally utilize the identical dataset evaluated by Worley et al. [5], who achieved state-of-the-art nominal performance on these constrained samples. By maintaining this strict 1:1 comparative baseline, we isolate the impact of adversarial perturbations and prove that generative augmentation successfully defends models previously thought to be secure. The measurements originate from the same experimental platform as Worley et al. [5], in which the ring oscillators are implemented on eight Digilent Nexys4 DDR boards built around a Xilinx Artix-7 (XC7A100T) FPGA fabricated at the 28 nm technology node; this identical RON frequency dataset has also been used in related on-chip-sensor and RON-based detection studies [31, 32]. We note that process variation scales with the technology node: at more advanced nodes, increased random dopant fluctuation and line-edge roughness widen the nominal spread of RO frequencies and may lower the signal-to-noise ratio between Trojan-induced shifts and process noise. Because the proposed defense does not assume any fixed variation magnitude, but instead learns and augments the empirical frequency manifold, it is expected to transfer to newer nodes once the synthesizers are retrained on node-specific golden measurements; direct validation on sub-28 nm silicon is nonetheless identified as important future work. For model evaluation, the data is randomly split into training

and testing sets using an 80:20 ratio. Feature normalization is performed using StandardScaler to ensure comparable feature scales and to optimize the performance of the SVM classifier.

3.3 Synthetic Sample Generation

Recent studies indicate that models trained on expanded datasets incorporating a broader range of diverse samples, including those generated by advanced generative architectures [33], demonstrate enhanced robustness to adversarial attacks. To mitigate data scarcity and to examine this defensive strategy within the context of hardware security, three data augmentation techniques are employed to generate synthetic training samples and to improve the resilience of hardware Trojan detection models.

These three techniques were selected because they represent three distinct and widely adopted paradigms for tabular data synthesis, which enables a controlled comparison of how the underlying generative mechanism, rather than merely the volume of added data, affects robustness. SMOTE is the canonical interpolation-based oversampler and provides a strong, non-parametric baseline. CTGAN and TVAE are the two leading deep generative models for heterogeneous tabular data, representing the adversarial (GAN) and variational (VAE) families, respectively; both employ mode-specific normalization tailored to tabular distributions, unlike image-oriented generators that perform poorly on low-dimensional continuous features. Purely image-domain models (e.g., DCGAN or diffusion models) and naive Gaussian/random oversampling were therefore excluded, as the former are ill-suited to eight-dimensional RON vectors and the latter cannot reproduce the strong inter-feature correlations that characterize RON side-channel signatures.

To ensure a rigorous and systematic evaluation of the proposed defense strategy, a structured data preparation and augmentation procedure was adopted. The original dataset, comprising 800 entries, was partitioned according to an 80/20 training and testing split. This process produced a highly imbalanced baseline training set of 640 samples, consisting of 588 Trojan inserted instances and 52 Trojan free instances. To establish our first defense baseline, SMOTE was applied to generate 536 synthetic Trojan-free samples, resulting in an exactly balanced training dataset of 1,176 samples (588 per class). For the proposed generative defense, TVAE was utilized to significantly expand manifold coverage. We generated 1,000 synthetic Trojan-inserted samples and 1,536 synthetic Trojan-free samples. When combined with the original training data, this produced a robust, perfectly balanced training dataset of 3,176 total samples (1,588 per class). Finally, for the comparative fidelity evaluation, CTGAN was trained under the same protocol to synthesize a comparable pool of samples. In all scenarios, the testing set (160 samples) remained completely isolated and strictly consisted of original, real-world measurements.

3.3.1 SMOTE (Synthetic Minority Oversampling Technique)

The dataset utilized in this study exhibits significant class imbalance, which can negatively impact learning efficiency and bias predictions toward the majority class [34]. To address this issue, we adopt the Synthetic Minority Oversampling Technique (SMOTE) [35]. Rather than simply duplicating minority-class samples, SMOTE generates new synthetic instances by interpolating between existing minority samples in the feature space.

Algorithm 1 Synthetic Minority Oversampling Technique (SMOTE).

Require: Minority class sample set S
Require: Oversampling rate N (%)
Require: Number of nearest neighbors k
Ensure: Set of generated synthetic minority samples S'
Initialize synthetic sample set $S' \leftarrow \emptyset$
for all samples $x_i \in S$ **do**
  Identify the k nearest neighbors of x_i within S
  for each synthetic sample required for x_i according to N **do**
    Randomly select a neighbor x_j from the k nearest neighbors
    Draw random interpolation factor $\lambda \sim \mathcal{U}(0, 1)$
    Generate synthetic sample: $x_{\text{new}} = x_i + \lambda \cdot (x_j - x_i)$
    Add x_{new} to S'
  end for
end for
return S'

Specifically, for a given minority-class sample, SMOTE identifies its k-nearest neighbors in feature space, and creates synthetic samples along the line segments connecting the sample to one of its neighbors. Each synthetic instance is generated by adding a scaled difference between the sample and its neighbor, where the scaling factor is randomly drawn from the interval [0, 1]. This interpolation-based strategy increases the diversity of minority-class samples while preserving the underlying data distribution. The detailed procedure is summarized in Algorithm 1.

3.3.2 TVAE (Tabular Variational Autoencoder)

The TVAE system uses variational autoencoders to discover data distribution patterns in complex datasets [36]. The system consists of two core components: an encoder that maps real data into a latent space and a decoder that reconstructs data samples from this latent space. By learning a structured latent distribution, the TVAE system is able to generate high-quality synthetic samples that preserve the statistical properties of the original data while exhibiting increased diversity.

TVAE is trained by optimizing a composite loss function composed of two essential parts:

- *Reconstruction Loss*: This loss measures the decoder's ability to accurately reproduce the original input data from the latent representation [37]. Mean Squared Error (MSE) is used for numerical/continuous features, while Binary Cross-Entropy (BCE) is applied to categorical features. The overall reconstruction loss is computed as the sum of these respective losses across all feature dimensions, corresponding to the negative log-likelihood of the data given the latent variables.
- *KL Divergence*: This second term regularizes the latent space by encouraging the learned posterior distribution to align with a standard normal distribution [38]. This is critical for preventing the model from simply memorizing the training data and instead forcing it to learn a smooth, continuous manifold from which diverse synthetic samples can be generated.

The total training objective is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{recon} + D_{KL}(q(\vec{z}|\vec{x})||p(\vec{z})) \quad (1)$$

where $\mathcal{L}_{recon}$ is the reconstruction loss and $D_{KL}(q(\vec{z}|\vec{x})||p(\vec{z}))$ is the Kullback-Leibler divergence between the learned distribution $q(\vec{z}|\vec{x})$ and the prior distribution $p(\vec{z})$. The detailed training procedure is summarized in Algorithm 2.

Algorithm 2 Training of TVAE for Synthetic Sample Generation

Require: Training dataset D
Require: Number of training epochs E
Ensure: Trained TVAE model for synthetic data generation
Initialize encoder network $Enc(\cdot)$ and decoder network $Dec(\cdot)$
for epoch $= 1$ to E **do**
  for all mini-batches $x \subset D$ **do**
    Encode input batch: $(\mu, \sigma) \leftarrow Enc(x)$
    Sample latent variable: $z \sim \mathcal{N}(\mu, \sigma)$
    Reconstruct data: $\hat{x} \leftarrow Dec(z)$
    Compute reconstruction loss $\mathcal{L}_{\text{recon}}(x, \hat{x})$
    Compute KL-divergence loss $D_{\text{KL}}\big(\mathcal{N}(\mu, \sigma) \,\|\, \mathcal{N}(0, I)\big)$
    Compute total loss: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{recon}} + D_{\text{KL}}$
    Update encoder and decoder parameters by minimizing $\mathcal{L}_{\text{total}}$
  end for
end for

3.4 Analysis Model

The Support Vector Machine (SVM) serves as the main classification model in this study due to its strong performance in prior hardware Trojan detection research. To identify the optimal model configuration, an exhaustive grid search is performed on the training set using "GridSearchCV". While advanced Bayesian optimization methods, such as the Tree-Structured Parzen Estimator (TPE) [39], have proven highly effective for hyperparameter tuning in complex neural and generative architectures, grid search is utilized here to ensure a transparent and comprehensive evaluation across the constrained parameter space of the SVM. The parameter grid includes three kernel options {"linear", "rbf", and "sigmoid"}, three regularization parameters $C \in \{0.1, 1, 10\}$, and three kernel scale options $\gamma \in \{0.01, 0.1,' scale'\}$. The grid search results indicate that the linear kernel with $C = 10$ and $\gamma =' scale'$ achieves the best overall performance across evaluation metrics. This configuration is therefore adopted as the baseline model for subsequent robustness and adversarial analysis. The value $C = 10$ was not set manually but emerged from the cross-validated search as the F1-maximizing configuration. Intuitively, the regularization parameter C governs the trade-off between margin width and training-error penalty; the moderately high value selected here reflects the penalty required under strong class imbalance, where a smaller C would widen the margin at the expense of misclassifying the scarce minority class. Larger penalties were not included, as the validation F1-score had already plateaued at $C = 10$, indicating diminishing returns and an increased risk of overfitting the constrained training set.

Algorithm 3 Gradient-Based Evasion Attack.

Require: Trained SVM classifier $model$
Require: Test feature matrix X_{test}
Require: True labels y_{test}
Require: Perturbation step size ϵ
Ensure: Perturbed test dataset X'_{test}
$X'_{\text{test}} \leftarrow$ copy of X_{test}
for all test samples $(x_i, y_i) \in (X_{\text{test}}, y_{\text{test}})$ **do**
  if $y_i = 1$ **then** ▷ Trojan-inserted sample
    Initialize gradient vector $\mathbf{g} \leftarrow \mathbf{0}$
    for feature index $j = 1$ to n **do**
      $x^+ \leftarrow$ copy of x_i
      $x^- \leftarrow$ copy of x_i
      $\delta \leftarrow 10^{-4}$
      $x_j^+ \leftarrow x_j^+ + \delta$
      $x_j^- \leftarrow x_j^- - \delta$
      $f^+ \leftarrow model$.decision_function(x^+)
      $f^- \leftarrow model$.decision_function(x^-)
      $g_j \leftarrow (f^+ - f^-)/(2\delta)$
    end for
    Normalize gradient direction: $\mathbf{d} \leftarrow \mathbf{g}/\|\mathbf{g}\|$
    Compute perturbation: $\Delta x \leftarrow -\epsilon \cdot \mathbf{d}$
    Update sample: $X'_{\text{test}}[i] \leftarrow x_i + \Delta x$
  end if
end for
return X'_{test}

3.5 Adversarial Attack Strategy

To assess the vulnerability of the trained SVM models, a feature-space adversarial attack is implemented to generate

the adversarially perturbed samples. The attack follows a gradient-based strategy that is applicable to SVMs with different kernel types, including linear, RBF, and sigmoid, without requiring direct access to model parameters. Instead, the gradient of the SVM decision function $f(\vec{x})$ is approximated numerically using finite-difference methods. The estimated gradient points toward the direction that most rapidly reduces the classifier's confidence in its prediction. The attacker exploits this direction to craft the adversarial perturbations that push the input sample toward the decision boundary. While mainstream adversarial attacks such as the Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD) are natively designed for differentiable deep neural networks via backpropagation, they cannot be directly applied to kernel-based SVMs. Therefore, the proposed approach serves as the mathematical and functional equivalent of an FGSM-style attack adapted for SVMs, utilizing numerical finite-difference estimation to accurately approximate the gradient and determine the optimal perturbation direction.

The gradient approximation for the *j*-th feature is calculated as:

$$\nabla f(\vec{x})_j \approx \frac{f(\vec{x}+\delta_j) - f(\vec{x}-\delta_j)}{2\delta} \tag{2}$$

where δ_j is a small perturbation applied along the *j*-th feature dimension and δ controls the finite-difference step size. The resulting adversarial example is then generated as:

$$\vec{x}_{adv} = \vec{x} - \epsilon \cdot \frac{\nabla f(\vec{x})}{||\nabla f(\vec{x})||} \tag{3}$$

where ϵ denotes the attack strength and constrains the magnitude of the perturbation. The full implementation details of the proposed gradient-based evasion attack are provided in Algorithm 3.

4 Experiment and Evaluation

4.1 Model Training

All experiments are conducted in a controlled environment running Python 3.10 and the *scikit-learn* library for classification tasks. Prior to training, all features are standardized to mitigate the effect of scale differences on SVM optimization. The dataset is partitioned into training and testing sets using an 80:20 split with stratified sampling to preserve the original class distribution.

SVM classifiers are trained using three kernel functions: linear, RBF, and sigmoid. Hyperparameter tuning is performed via grid search with three-fold cross-validation. The regularization parameter C is evaluated over $\{0.1, 1, 10\}$, while the kernel parameter γ for nonlinear kernels is selected from $\{0.01, 0.1, 1, scale\}$. Model selection is based on maximizing the F1-score on the validation sets, as this metric effectively balances precision and recall in class-imbalanced detection scenarios.

For data augmentation experiments, the SMOTE algorithm is applied to generate synthetic Trojan-free (minority-class) samples until the minority and majority classes are balanced. In addition, generative models from the SDV library, specifically TVAE and CTGAN, are trained for up to 2,500 epochs with a batch size of 128 [40]. To mitigate the potential overfitting risks associated with extended generative training, early stopping is enforced by continuously monitoring the validation reconstruction loss. Furthermore, extracting training data from eight distinct FPGA boards inherently introduces hardware process variation. Rather than degrading performance, this variance acts as a natural regularizer, ensuring the TVAE maps a generalized latent space instead of memorizing board-specific noise. The dataset's distribution across 32 independent physical zones provides sufficient statistical power and diversity to robustly validate the classifier's decision boundaries.

For the adversarial-training baseline, the adversarial samples are produced by applying the finite-difference gradient evasion attack of Algorithm 3 to the Trojan-inserted instances of the training partition, using the same perturbation budget adopted at evaluation time. These perturbed Trojan-inserted samples (588 in total) retain their original "Trojan-inserted" labels and are appended to the 640-sample clean training set, yielding an augmented set of 1,228 samples in which adversarial examples constitute approximately 48% of the data. The classifier is then retrained on this augmented set following the identical grid-search procedure, so that the only difference from the baseline is the presence of adversarial samples during training. This design isolates the effect of adversarial training and enables a like-for-like comparison against the generative-augmentation defenses.

4.2 Evaluation Protocol

The evaluation protocol is designed to assess two complementary aspects: the fidelity of generated data and the robustness of classification models under adversarial conditions.

4.2.1 Data Generation Evaluation

SDMetrics evaluate the accuracy of synthetic data through its metrics [41]: the *column shapes score* evaluates how well synthetic data features match real data

features, the *column pair trends* score measures how well synthetic data maintained real data correlations, and the *diversity score* measures the number of distinct synthetic records relative to the total number of generated records for robustness evaluation. The evaluation of synthetic data requires both realistic and diverse characteristics, which these metrics assess to determine their suitability for improving classifier performance.

4.2.2 Classification Metrics

Classifier performance is measured using accuracy, precision, recall, F1-score, false positive rate (FPR), and false negative rate (FNR) [42]. These metrics are derived from the confusion matrix, where *TP*, *TN*, *FP*, and *FN* denote true positives, true negatives, false positives, and false negatives, respectively. The definitions of these metrics are given as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \tag{5}$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{6}$$

$$\text{FPR} = \frac{FP}{FP + TN}, \quad \text{FNR} = \frac{FN}{FN + TP} \tag{7}$$

Among these metrics, recall is given particular emphasis, as a reduction in recall corresponds to Trojan-inserted circuits being misclassified as Trojan-free, which represents the most critical failure mode in practical hardware security scenarios.

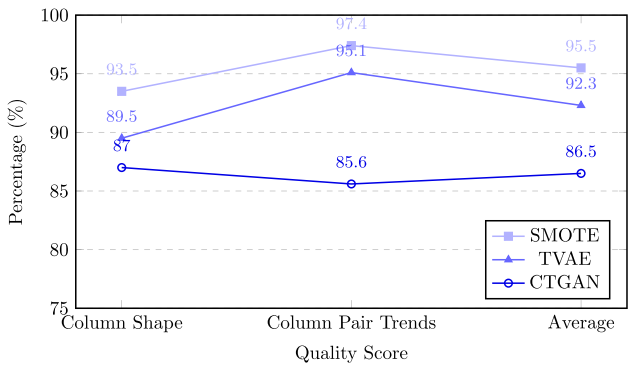


Fig. 3 Synthetic data fidelity comparison across SMOTE, CTGAN, and TVAE using SDMetrics scores

4.3 Adversarial Evaluation

Classifier robustness is evaluated under two adversarial attack settings. For linear SVMs, the explicit form of the decision boundary enables direct feature-space attacks by applying perturbations along the direction orthogonal to the separating hyperplane. In this setting, Trojan-inserted samples are iteratively perturbed according to the update rule until they cross the decision boundary. For more general cases, the gradient of the SVM decision function is approximated using numerical central finite-difference estimation. The attack progresses until the classifier's confidence is reduced to zero, at which point the perturbed sample is misclassified into the Trojan-free decision region.

While deep learning architectures, such as Convolutional Neural Networks (CNNs), are increasingly applied to side-channel analysis, this study limits its experimental scope to SVM classifiers. This choice is deliberate; prior comparative evaluations on this exact dataset established SVMs as one of the highest-performing models for hardware Trojan detection [5]. Therefore, demonstrating adversarial vulnerabilities and evaluating generative defenses on a formally recognized, best-in-class baseline provides a more rigorous and practically relevant assessment than introducing deep learning models, which carry a high risk of overfitting on tabular, constrained RON data.

All experiments are conducted using a fixed random seed to ensure reproducibility. To maintain physical realism, perturbation magnitudes are constrained to remain within 5% of each feature's dynamic range. This bound is empirically justified by typical hardware operational tolerances; it reflects plausible side-channel variations induced by ambient temperature drift and supply voltage (V_{DD}) noise, ensuring the adversarial manipulations remain physically viable and indistinguishable from natural environmental fluctuations. Consistent with the assumed threat model, the attacker is unable to apply large or overt hardware modifications, as such changes would be detectable through physical inspection or validation procedures.

The physical plausibility of the synthetic samples themselves is likewise safeguarded. The SDMetrics column-shape and column-pair-trend scores (Fig. 3) quantify the agreement between the marginal distributions and the inter-RO correlations of the synthetic and real measurements, while the PCA projections (Figs. 4–8) confirm that the generated RO-frequency vectors occupy the same region of feature space as the real data. Because the generative models are trained directly on measurements acquired across eight physical boards and 32 independent zones, the natural process-, temperature-, and supply-voltage-induced variability present during real acquisition is already embedded in the training distribution and, consequently, in the synthesized

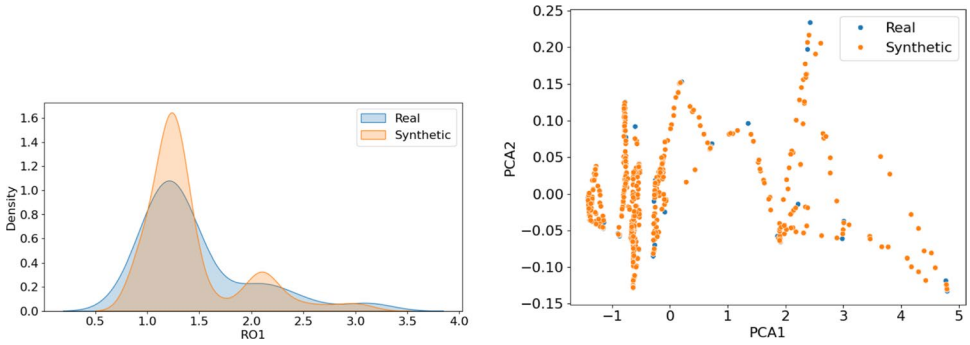


Fig. 4 Comparison of SMOTE data augmentation results: (**a**) distribution comparison for RO1 Trojan-free class; (**b**) real versus synthetic data in PCA space

vectors. A dedicated validation against freshly acquired silicon measurements collected under controlled temperature and V_{DD} sweeps would further reinforce this evidence and is identified as future work.

Evaluation is performed under two testing conditions: (i) normal operation without adversarial manipulation and (ii) feature-space adversarial attacks. For each model configuration, experiments are conducted using the baseline dataset as well as datasets augmented with SMOTE and TVAE-generated samples. Model robustness is quantified by measuring the degradation in detection performance when transitioning from benign to adversarial conditions, with larger performance drops indicating greater susceptibility to evasion attacks.

5 Results and Analysis

The experimental evaluation focused on three primary areas: (i) the fidelity and distributional realism of synthetic data generated using SMOTE, CTGAN, and TVAE; (ii) the nominal, no-attack performance of Support Vector Machine (SVM) classifiers trained under different defense strategies; and (iii) the robustness of these classifiers when subjected to adversarial evasion attacks using both linear feature-space perturbations and gradient-based attacks. Each defense mechanism—including classical adversarial training and synthetic-data-driven augmentation—was assessed under clean and adversarially perturbed conditions to quantify improvements in adversarial resilience.

5.1 Quality of Synthetic Samples

The efficacy of the synthetic data generation models was formally assessed using the SDMetrics framework, evaluating both column shape similarity and column pair trends [41].

As illustrated in Fig. 3, all three techniques preserved the statistical structure of the real RON data, but with a distinct fidelity–diversity trade-off. SMOTE attained the highest marginal fidelity—a column shape similarity of 93.5%, a column pair trend similarity of 97.5%, and an average quality score of 95.5%—which is expected, since linear interpolation between real minority samples reproduces the observed marginal distributions almost exactly. The Tabular Variational Autoencoder (TVAE) achieved slightly lower yet still high fidelity, with a column shape similarity of 89.5%, a column pair trend similarity of 95.1%, and an average score of 92.3%, together with a near-maximal sample-diversity score of approximately 1.0, which confirms that it generated genuinely novel samples rather than memorizing the training data. CTGAN exhibited the lowest fidelity (column shape 87.0%, column pair trend 85.6%, average 86.5%), particularly in preserving complex inter-feature correlations. Although SMOTE and TVAE report comparable numeric diversity, the PCA projections presented in the following subsections reveal a fundamental geometric difference: SMOTE samples remain confined to linear segments within the convex hull of the real data, whereas TVAE populates the non-linear regions of the underlying manifold. As detailed in Section 5.3, it is this manifold coverage, rather than raw marginal fidelity, that governs adversarial robustness and motivates the selection of TVAE.

5.1.1 SMOTE: Linear Interpolation-Based Sample Expansion

SMOTE balances the dataset by interpolating between minority samples. While effective for simple upsampling, its limitation lies in its inability to generate data outside the convex hull of existing points.

Figure 4 shows the distribution and PCA visualization for SMOTE. The PCA plot (Fig. 4b) reveals that synthetic samples (orange) lie strictly along the linear paths between real samples (blue), confirming that SMOTE does not expand the decision boundary into the non-linear space required to defend against complex adversarial perturbations.

5.1.2 TVAE: Latent-Space Generative Sample Modeling

TVAE utilizes a variational autoencoder to learn the underlying probability distribution of the data.

As shown in Figs. 5 and 6, TVAE synthetic samples closely follow the density of the real data while introducing necessary variance. The PCA visualizations demonstrate that TVAE covers the feature space more comprehensively than SMOTE, filling gaps in the data manifold that are critical for robust decision boundary formation.

5.1.3 CTGAN: Adversarial Modeling of Tabular Distributions

CTGAN uses GAN-based training to synthesize tabular data. Figures 7 and 8 indicate that while CTGAN captures the general data distribution, it exhibits more scatter and slightly lower overlap with high-density regions of the real

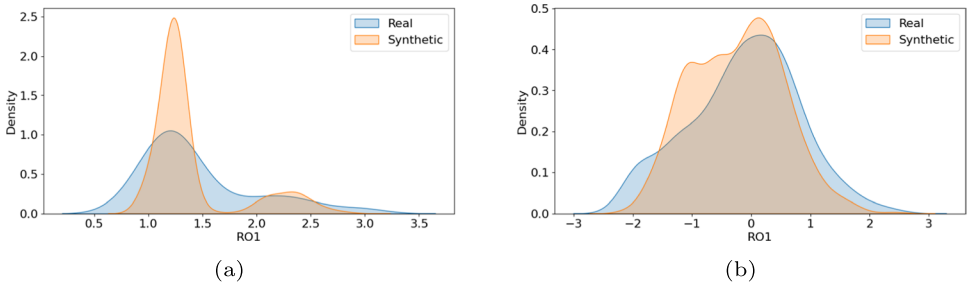


Fig. 5 Distribution comparison of TVAE-augmented RO1 data: (**a**) Trojan-free class; (**b**) Trojan-inserted class

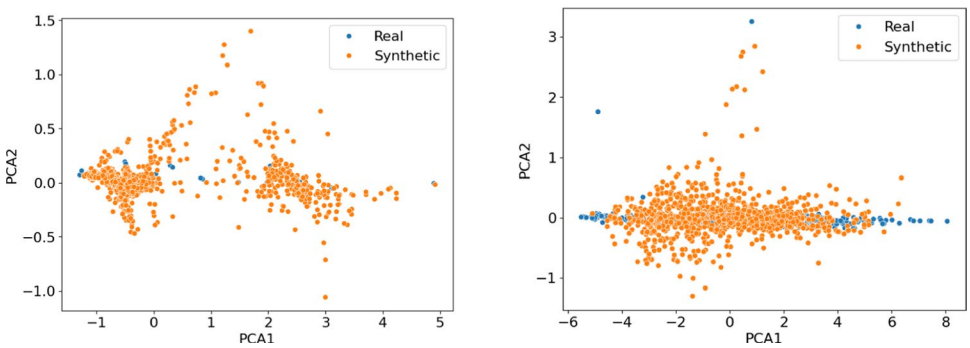


Fig. 6 PCA-based visualization of real and synthetic RO1 samples generated using TVAE: (**a**) Trojan-free class; (**b**) Trojan-inserted class

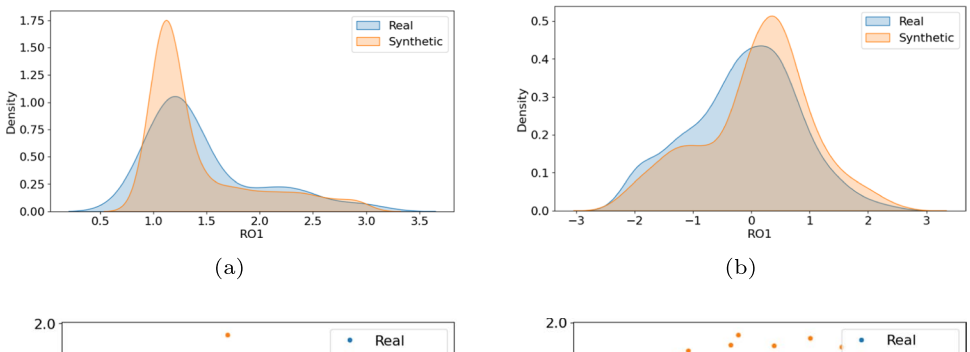


Fig. 7 Distribution comparison of CTGAN-augmented RO1 data: (**a**) Trojan-free class; (**b**) Trojan-inserted class

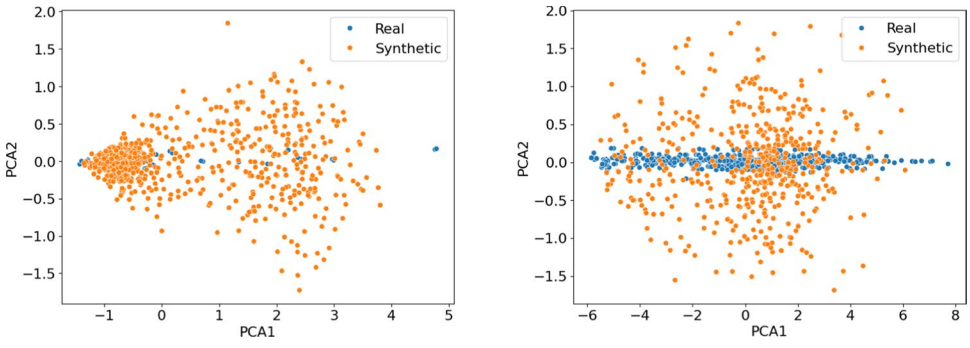


Fig. 8 PCA-based comparison of real and synthetic RO1 samples generated using CTGAN: (**a**) Trojan-free class; (**b**) Trojan-inserted class

data compared to TVAE. Quantitative analysis indicates that the distinctive properties of the synthetic data generated by TVAE, particularly its 95.1% inter-feature (column pair trend) similarity combined with a near-maximal (≈ 1.0) sample diversity, contribute directly to the formation of a more robust decision boundary. In contrast, the comparatively lower fidelity of CTGAN results in dispersed samples that may introduce noise and irregularities into the boundary construction. The structured diversity provided by TVAE more effectively represents the non linear regions of the underlying hardware manifold. As discussed in the adversarial evaluation presented in Section 5.3, this mechanism influences the selection of support vectors in the SVM classifier, leading to the establishment of a wider and more conservative geometric margin. On this basis, TVAE was selected over CTGAN for training the robust SVM model used in the adversarial evaluation.

5.2 Nominal Performance of SVM Classifiers (Before Attack)

The baseline performance of the SVM classifiers was first evaluated under benign, no-attack conditions to ensure that defense mechanisms did not degrade standard detection capabilities.

Figures 9 and 11a summarize the results. The baseline model (None) achieved high recall (1.0) but suffered from a high False Positive Rate (FPR) of 0.6667, a known issue in RON-based detection where noise is often mistaken for Trojan activity. In a practical semiconductor manufacturing setting, such a high FPR is economically untenable, as it leads to the unnecessary rejection of functional integrated circuits, severely degrading production yield and escalating manufacturing costs. TVAE-augmented training significantly improved the balance between precision and recall, achieving an F1-score of over 95% and lowering the FPR to 0.0%. Standard adversarial training improved recall but resulted in a higher FPR compared to the generative approaches (Fig. 10).

5.3 Performance Under Adversarial Evasion Attacks

The critical evaluation of this study focuses on model robustness against adversarial evasion attacks. As demonstrated in Fig. 11b and the confusion matrices in Fig. 10, the baseline SVM model is catastrophically vulnerable to gradient-based attacks. To systematically analyze the impact of perturbation intensity, a sensitivity evaluation was conducted on the baseline SVM across varying attack step sizes (ϵ), ranging from minimal noise ($\epsilon = 0.01$) to the maximum physical constraint ($\epsilon = 1.0$). As illustrated in Fig. 12, the baseline model robustly retains high recall and accuracy under minor perturbations. However, as the attack intensity scales, the geometric fragility of the decision boundary is exposed. At a moderate intensity ($\epsilon = 0.33$), recall degrades sharply to 61.49%, and at the maximum threshold ($\epsilon = 1.0$), recall

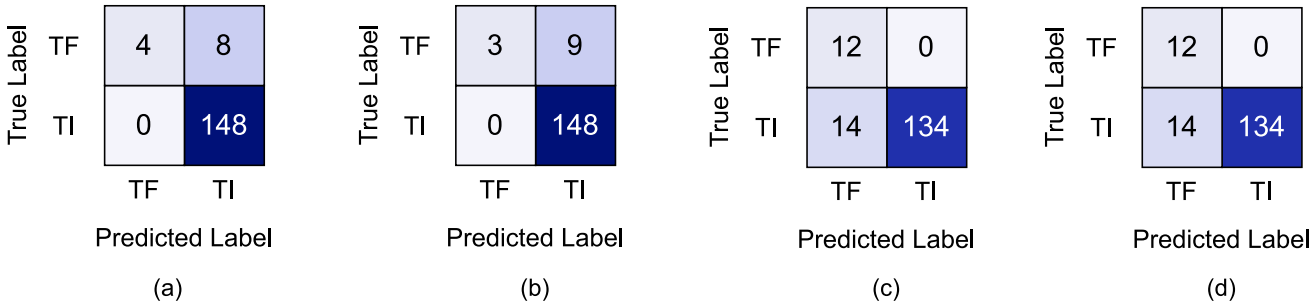


Fig. 9 Confusion matrix for SVM models with different defense strategies with clean test data. (**a**) None, (**b**) Adversarial Training, (**c**) SMOTE, (**d**) TVAE

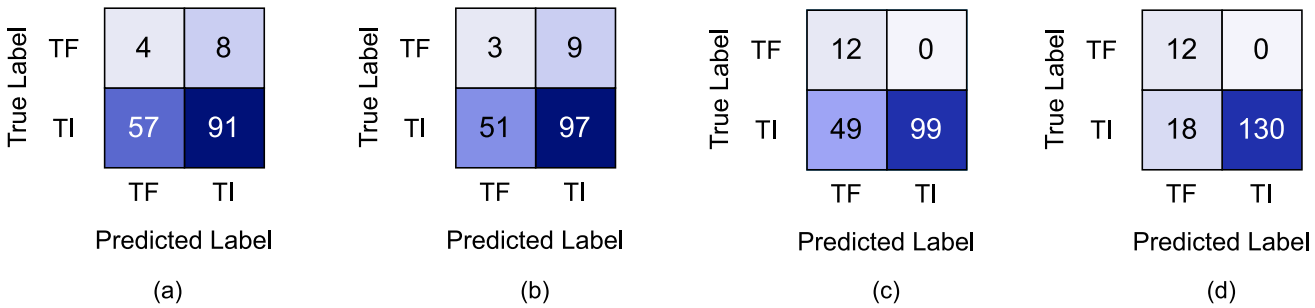


Fig. 10 Confusion matrix for SVM models with different defense strategies with adversarially perturbed test data. (**a**) None, (**b**) Adversarial Training, (**c**) SMOTE, (**d**) TVAE

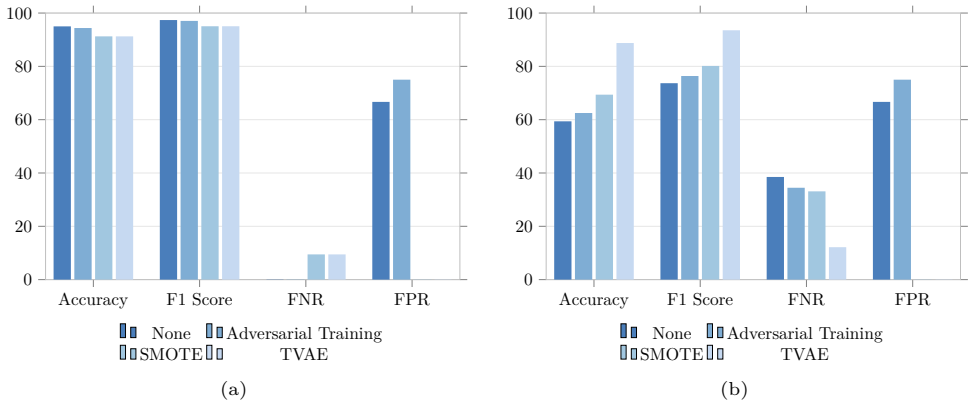


Fig. 11 Performance metrics for evaluating different defense strategies: (**a**) clean test data; (**b**) adversarially perturbed test data

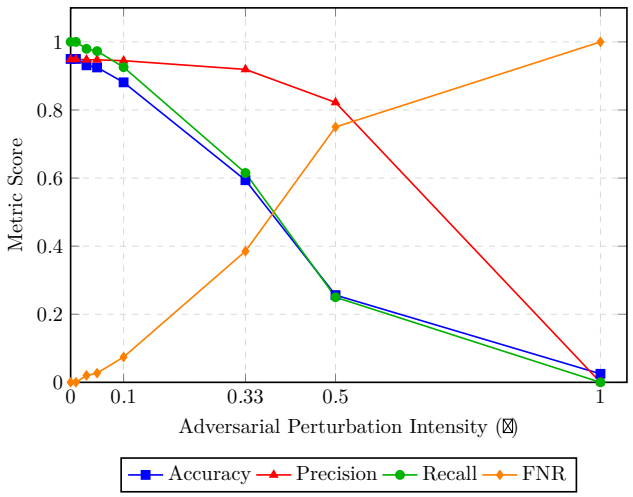


Fig. 12 Sensitivity analysis showing the degradation of baseline SVM performance metrics under varying adversarial perturbation intensities (ϵ)

completely collapses to 0.0%, with a corresponding spike in the False Negative Rate (FNR) to 1.0. This systematic degradation curve visually confirms that the baseline model is highly susceptible to targeted gradient descent once the perturbation surpasses ambient noise levels.

SMOTE provided moderate improvements, achieving a recall of 66.89% under gradient attack. In contrast, the TVAE-augmented model demonstrated superior robustness. Even under strong gradient-based perturbations, the TVAE model maintained a recall of 87.84% and an overall accuracy of 88.75%. Because both SMOTE and TVAE achieve exact class balance during training, this stark difference in robustness analytically demonstrates that the resilience of TVAE stems from enhanced manifold coverage rather than mere class balancing. By synthesizing diverse samples that populate the non-linear gaps within the minority class distribution (as corroborated by the PCA visualizations), TVAE alters the distribution of the SVM's support vectors. This forces the optimization algorithm to construct a wider, more conservative geometric margin, effectively pushing the decision boundary further from the true Trojan-inserted data manifold and requiring the adversary to apply significantly larger perturbations to successfully induce misclassification. The confusion matrix in Fig. 10(d) confirms that TVAE successfully prevented the majority of evasion attempts, correctly identifying 130 out of 148 Trojan instances that were missed by the baseline model. This establishes TVAE-based synthetic data augmentation as a highly effective defense strategy for securing hardware Trojan detection systems. A natural question is whether the same robustness could instead be obtained simply by collecting more diverse real measurements across additional boards and operating conditions. In principle it could, since broader real coverage would similarly densify the feature manifold; in practice, however, acquiring, instrumenting, and characterizing large numbers of golden and Trojan-inserted chips across many operating points is costly and often infeasible in a screening pipeline. TVAE offers a low-cost proxy for this effect: it learns the empirical manifold already spanned by the eight boards and 32 zones and interpolates the non-linear regions between observed operating points, thereby delivering much of the robustness benefit of additional real data without the associated fabrication and measurement burden. The controlled comparison with SMOTE, which receives the identical class balance but only linear coverage, isolates this manifold-densification effect as the operative mechanism rather than data volume alone.

5.4 Robustness Across Multiple Data Splits

To verify that these findings are not artifacts of a single train/test partition, the evaluation was repeated over 20 independent stratified random 80/20 splits for all three configurations—the baseline SVM, the SMOTE-augmented SVM, and the proposed TVAE-augmented SVM—as summarized in Fig. 13. Nominal (clean) recall was stable across splits for every model (baseline $99.0\% \pm 1.6\%$, SMOTE $87.0\% \pm 2.3\%$, TVAE $87.8\% \pm 2.6\%$), and a complementary 5×5-fold cross-validation confirmed the baseline's clean

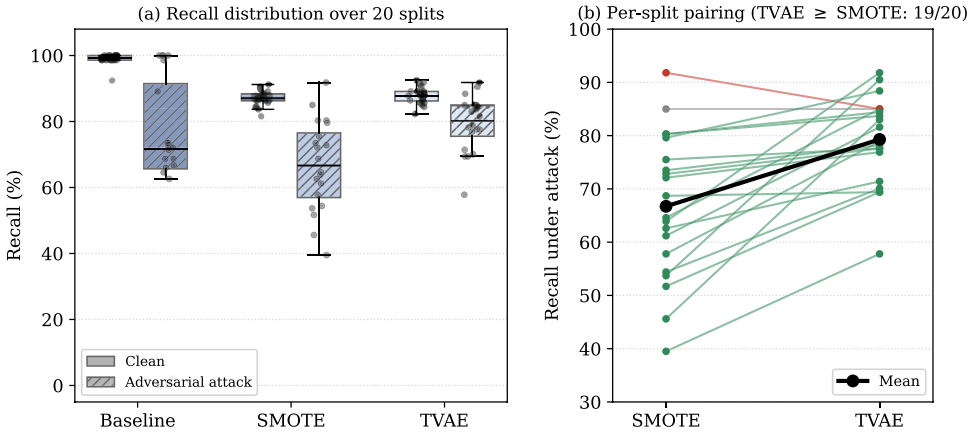


Fig. 13 Robustness across 20 random data splits: (**a**) recall distributions under clean and adversarial conditions; (**b**) per-split paired comparison of adversarial recall (SMOTE vs. TVAE)

accuracy at $91.4\% \pm 1.4\%$, indicating that the reported nominal performance is not split-dependent. Under the transferred gradient attack, the undefended SVM degraded severely and erratically, its mean recall falling to $67.6\% \pm 31.7\%$ and collapsing to near-zero on the most vulnerable splits. Class balancing alone (SMOTE) provided only limited and unstable protection ($66.7\% \pm 13.7\%$), whereas the TVAE-augmented model retained a substantially higher and far more stable recall of $79.3\% \pm 8.4\%$. As Fig. 13(a) shows, the undefended baseline degrades widely and erratically under attack, SMOTE remains insufficient, and TVAE preserves a higher and much tighter recall band. Importantly, this advantage is also consistent at the level of individual splits. In the paired comparison of Fig. 13(b), each line joins the SMOTE and TVAE adversarial recall for the same split; all but one line is non-decreasing, indicating that TVAE achieved higher adversarial recall than SMOTE on 18 of the 20 splits and matched it on one further split. The robustness gain is therefore systematic rather than an aggregate artifact. The primary $80:20$ split reported in Table 1 (87.8% recall) lies in the upper range of this distribution, and even at its lower end the TVAE defense preserves a decisive margin over both the balancing-only and the undefended models. These results establish that the vulnerability of the undefended detector, the insufficiency of a balancing-only defense, and the manifold-coverage advantage of the generative defense all persist across repeated sampling.

To explicitly benchmark our defense against prior state-of-the-art results, Table 1 presents a direct comparison with the baseline metrics from Worley et al. [5]. While the original models achieve high nominal accuracy, our surrogate model's performance collapses under feature-space attacks (FSA). Incorporating TVAE not only recovers the True Positive Rate (TPR) to 0.878 under severe attack but also optimally reduces the False Positive Rate (FPR) to 0.000, demonstrating robust defense capabilities on highly constrained tabular data.

6 Conclusion

This work presented a comprehensive evaluation of the vulnerability of machine learning-based Hardware Trojan detection systems to adversarial evasion attacks and proposed a generative defense framework. Our experimental results confirm that while Support Vector Machines based on Ring Oscillator Network data achieve high nominal accuracy, they are brittle against adversarial perturbations, with standard models suffering a complete collapse in recall (dropping to 0%) under gradient-based attacks. To address this, we explored synthetic data augmentation using

Table 1 Performance comparison of the proposed defense against prior baseline models on the referenced dataset. Bold values highlight the results of the proposed method

Model	TNR	FPR	FNR	TPR	Accuracy	Attack Applied?	Defense Strategy
SVM [5]	0.929	0.072	0.023	0.977	0.974	X	X
SVM+KNN+NB Ensemble [5]	0.908	0.092	0.055	0.945	0.943	X	X
SVM surrogate model [5]	0.333	0.667	0.385	0.615	0.594	FSA	X
SVM	0.250	0.750	0.345	0.655	0.625	FSA	Adversarial training
SVM (Proposed)	**1.000**	**0.000**	**0.095**	**0.905**	**0.913**	**X**	**TVAE**
SVM (Proposed)	**1.000**	**0.000**	**0.122**	**0.878**	**0.888**	**FSA**	**TVAE**

SMOTE, CTGAN, and TVAE. Among these, the Tabular Variational Autoencoder (TVAE) proved to be the most effective. TVAE generated high-fidelity synthetic samples that accurately captured the complex correlations of the side-channel data. Incorporating these samples into the training process significantly hardened the classifier. The TVAE-augmented model maintained over 91% accuracy on clean data and, most critically, retained approximately 88% accuracy and recall under severe adversarial attacks that defeated all other methods. These findings highlight that high nominal accuracy is insufficient for hardware security and establish generative data augmentation as a vital tool for building adversarially robust HT detectors.

A limitation of the current empirical evaluation is its exclusive focus on ring oscillator (RO) frequency measurements. We acknowledge that RON-based sensing has inherent detection limits: because it observes Trojans indirectly through their impact on the local power and timing environment, ultra-small or dormant Trojans whose switching activity does not measurably perturb nearby oscillators can remain below the detection threshold, and RON coverage depends on oscillator placement relative to the payload. Overcoming these blind spots generally requires complementary side channels rather than a different classifier. However, the proposed TVAE-augmented defense framework is fundamentally modality-agnostic, and this is precisely what allows it to move beyond the RON setting. It can be naturally extended to secure other high-fidelity sensor modalities, such as voltage-droop monitors designed to detect timing fault attacks induced by hardware Trojans [15]. Because generative synthetic augmentation operates on the underlying data distribution rather than specific hardware features, applying TVAE to these alternative side-channel signatures represents a highly promising avenue for broadening adversarial resilience across diverse hardware security deployments. To complement this expansion into new sensor modalities, future work will also evaluate the generalizability of TVAE-augmented defenses across other state-of-the-art algorithmic architectures, including ensemble methods such as XGBoost and deep learning classifiers.

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Author Contributions A.G. conceptualized the study, developed the methodology and software, conducted the formal analysis and investigation, wrote the original draft, and contributed to visualization and validation. L.C. contributed to conceptualization, validation, review and editing, and provided research planning insights and conceptual guidance. M.T.R. was responsible for data acquisition, hardware-level experiments, RO setup, and resources, and contributed to validation. C.C. and R.D. contributed to validation, experimental support, preliminary result verification, and interpretation discussions. F.A. contributed to conceptualization, review and editing, provided resources, supervised the project, handled project administration, and acquired funding. All authors have read and agreed to the published version of the manuscript.

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Data Availability The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Materials Availability Not applicable.

Code Availability Not applicable.

Declarations

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Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

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